Australian National University
Research School of Earth Sciences

Groundwater -- Principles

EMSC3025/6025

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Objectives

By the end of this module, you will be able to:

- **Understand porosity**: Calculate and interpret porosity in different geological materials
- **Apply Darcy's Law**: Use the fundamental equation to analyze groundwater flow
- **Measure hydraulic properties**: Distinguish between hydraulic head, conductivity, and permeability
- **Analyze flow systems**: Interpret flow patterns in heterogeneous and anisotropic media
- **Use field methods**: Design monitoring networks using wells and piezometers
- **Create flow maps**: Construct potentiometric surfaces and interpret hydrographs
- **Solve practical problems**: Apply groundwater principles to real-world scenarios

Groundwater Flow Theory

- This module introduces the principal laws that govern groundwater movement through porous media.
- We will explore:
  - Definitions and significance of **porosity**
  - **Darcy's Law**: the foundational equation for flow in saturated zones.
  - Concepts of **hydraulic head**, **conductivity**, and **velocity**
- These principles form the basis for:
  - Understanding aquifer behavior
  - Designing wells and evaluating sustainability
  - Developing numerical models

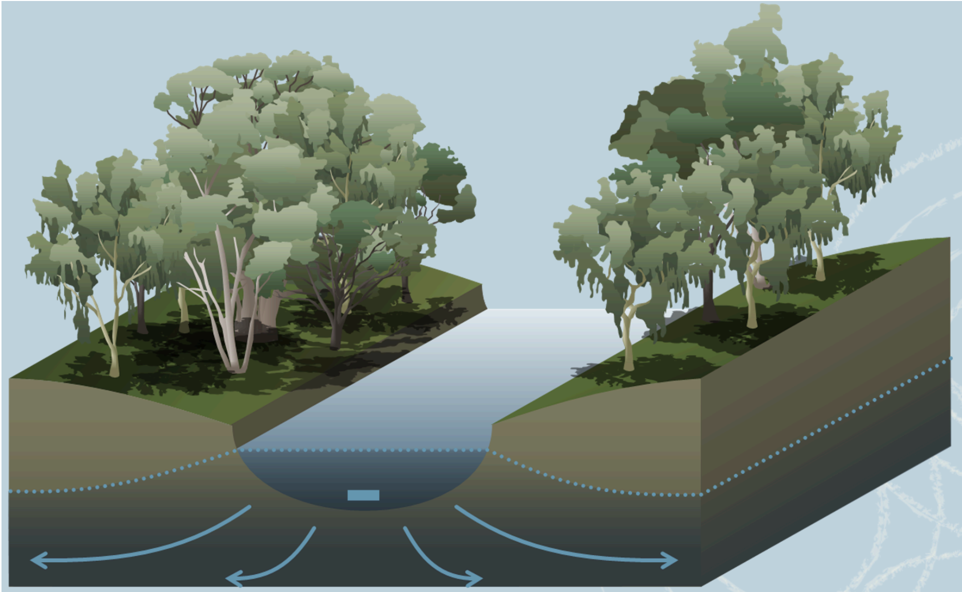

Figure 1: From wetlands to aquifers, we will explore the principles of groundwater flow.

Porosity of a Soil or Rock

- **Porosity** describes how much void space exists in a rock or soil—essentially, how much room is available to hold water.
- Groundwater resides in the pores of rocks and sediments.
- **Total porosity** n_T is defined as:

$$n_T = \frac{V_v}{V_T} = \frac{V_T - V_s}{V_T}$$

where:

  - V_v = volume of voids
  - V_s = volume of solids
  - V_T = total volume

- Typical values of n_T:
  - Sand and gravel: 0.25 – 0.35
  - Cemented sandstones: 0.05 – 0.15

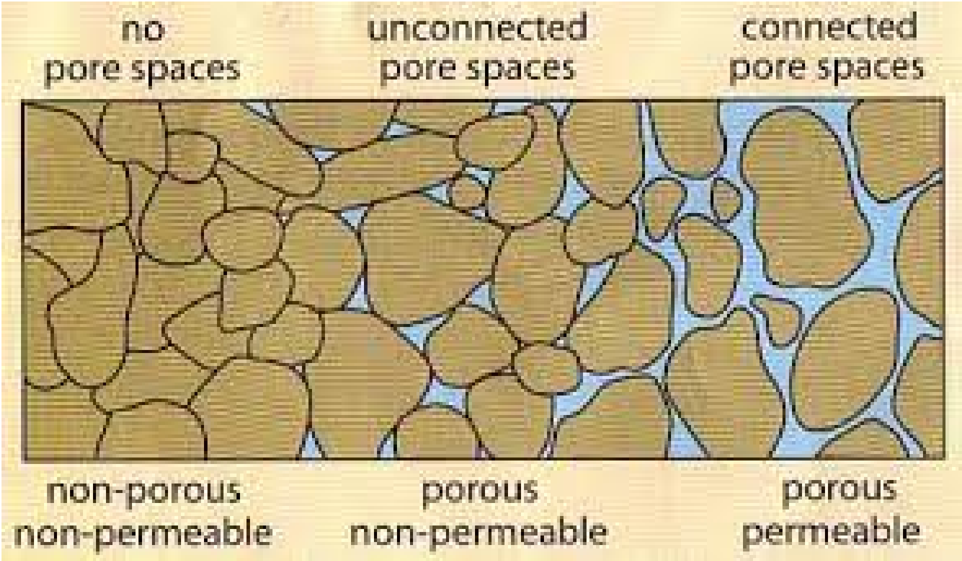

Figure 2: Porosity and permeability are connected, but not the same.

Porosity and Bulk Density

- Porosity can also be calculated using **bulk density**:

$$n_T = 1 - \frac{\rho_b}{\rho_s}$$

where:

  - ρ_b = bulk density (mass/volume of dry sample)
  - ρ_s = particle density (mass/volume of solids)

- This approach is common in lab settings where volume and mass can be measured directly.

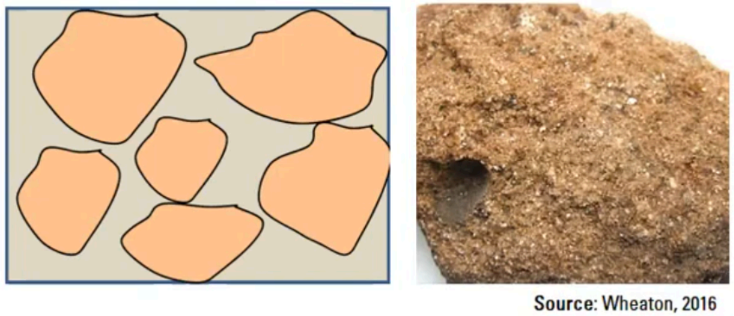

Figure 3: Primary vs Secondary Porosity

It is also useful to distinguish:

- **Primary porosity**: original voids formed during rock/soil formation (e.g., in sediments, vesicles in lava).
- **Secondary porosity**: formed later, e.g., via fractures, faults, or dissolution in carbonates.

Factors Affecting Porosity

- **Primary porosity** depends on:
  - **Compaction**: tighter packing = lower porosity
  - **Grain shape and arrangement**: more spherical = less compact = higher porosity
  - **Sorting**:
    - Well-sorted sediments (uniform size) → higher porosity
    - Poorly sorted (mixed sizes) → smaller grains fill the voids
- **Porosity ranges**:
  - 0 to >50% depending on material
  - Unlithified materials (e.g. sand) >> lithified ones (e.g. sandstone)

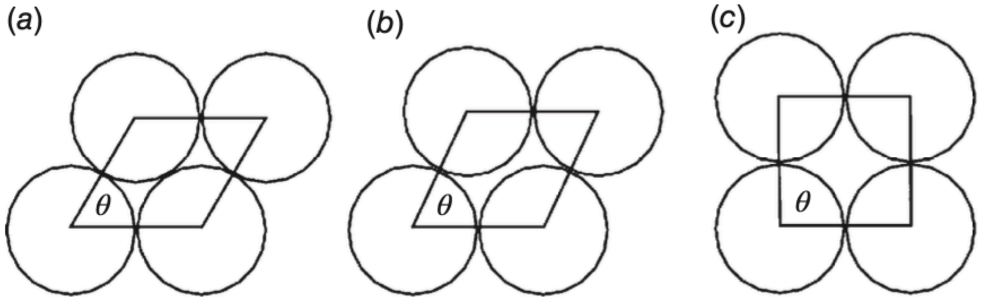

Figure 4: Sections of four contiguous spheres of equal size: (a) the most compact arrangement, lowest porosity; (b) less compact arrangement, higher porosity; (c) least compact arrangement, highest porosity

Good sorting and less compaction = more room for groundwater.

Porosity in Natural Materials

- For **unlithified materials**:
  - **Smaller grain size** → higher porosity (e.g. clays > sands)
  - Glacial till is variable (mixtures of sand/silt/clay + varying compaction)
- **Sandstone**:
  - Porosity reduced by cementation
- **Carbonates (e.g., limestone, dolomite)**:
  - Diagenesis and reactivity make porosity highly variable
- **Igneous rocks**: low primary porosity
- **Fractured rocks**:
  - Have **dual porosity**: matrix + fractures
  - Fractures provide **secondary porosity** essential for flow

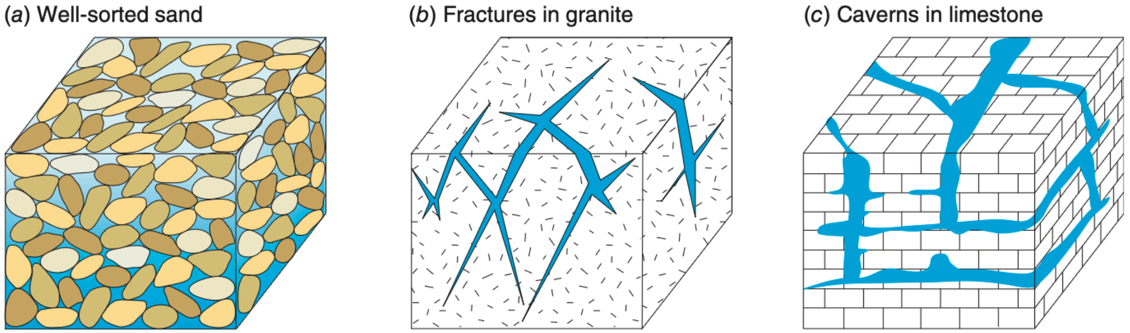

Figure 5: Types of openings in selected water-bearing rocks. Block (a) is several millimeters wide and shows primary porosity for unlithified sand. Blocks (b, c) are larger potential meters to 10s of meters in width. For granite (b), secondary porosity created by fracturing facilitates flow. Block (c) illustrates caverns in limestone produced by solution-enhanced enlargements of fractures.

Never judge porosity by rock type alone—structure and history matter.

Effective Porosity and Dual Porosity Systems

- **Effective porosity**: fraction of pore space that actually contributes to flow
  - Often < total porosity (Not all the pores are connected)
  - Especially important in fractured or poorly connected media
- **Fractured media** (e.g. shales, granite):
  - Matrix may store water, but fractures control flow
  - Fracture porosity n_f for shale: ~0.05 or less
- **Dual porosity** systems:
  - Primary: matrix porosity
  - Secondary: fractures, dissolution features (e.g. caverns in limestone)
- Porosity must be paired with **permeability** (up next!) to understand flow.

Figure 6: Outcrops of Coconino Sanstone along Silver Creek in northeastern Arizona. Notice the fracturing in vertical and horizontal directions.

Range of Porosity values

Material	Min	Mean	Max	SD
Unlithified deposits				
Sand	28.1	38.9	44.4	4.9
Dune	c	42.1	c	8.3
Silt	31.5	45.2	50.8	5.6
Clay	40.1	46.1	55	4.5
Glacial till	14.3	23.5	34	c
Peat	c	92	c	c

Table 1

Material	Min	Mean	Max	SD
Sedimentary rocks				
Sandstone	4	20.4	28.6	8.6
Limestone	0.44	8.67	34.7	8.6
Dolostone	0.4	7.46	26	c
Shale	0.75	10.8	27.2	8.3
Igneous rocks				
Granite	<0.001		0.01	
Basalt	<0.01		0.1	

Table 2

Occurrence of Groundwater

- Water in the subsurface occupies **pores** in soil and rock.
- **Unsaturated zone**: pores contain both air and water
  - Water content fluctuates with infiltration and evapotranspiration
  - Plants use this water
- **Saturated zone**: pores are fully water-filled — this is **groundwater**
- The **water table** separates the two zones
- A small portion of rain infiltrates deep enough to reach the water table → **recharge**

Water in the ground isn't just "there"—it's dynamic and layered.

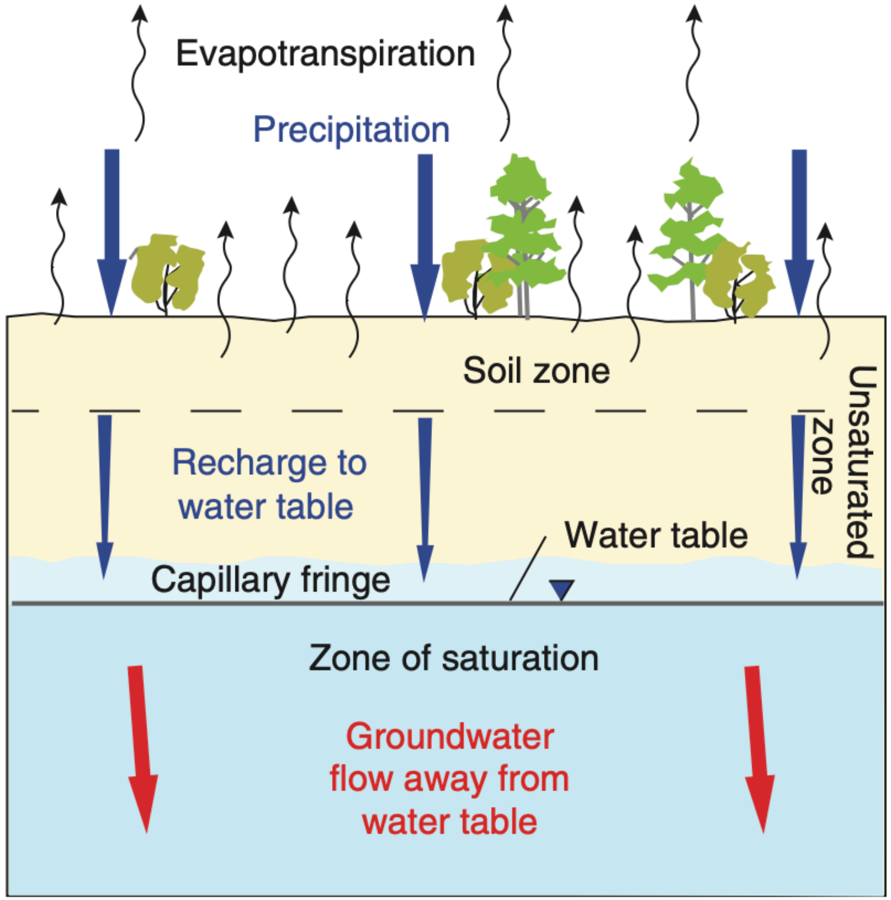

Figure 7: Recharge, unsaturated zone, and water table

Flow of Groundwater

- After recharge, groundwater flows:
  - **Downward** from water table due to gravity
  - **Sideways** following pressure gradients, often toward streams
- This flow forms **discharge** zones (e.g. rivers, springs)
- Example: Rain → infiltration → percolation → river discharge
- **Timescales vary**:
  - Surface runoff = hours
  - Groundwater discharge = months to years
- Flow influenced by:
  - Water table shape
  - Conductivity of subsurface materials

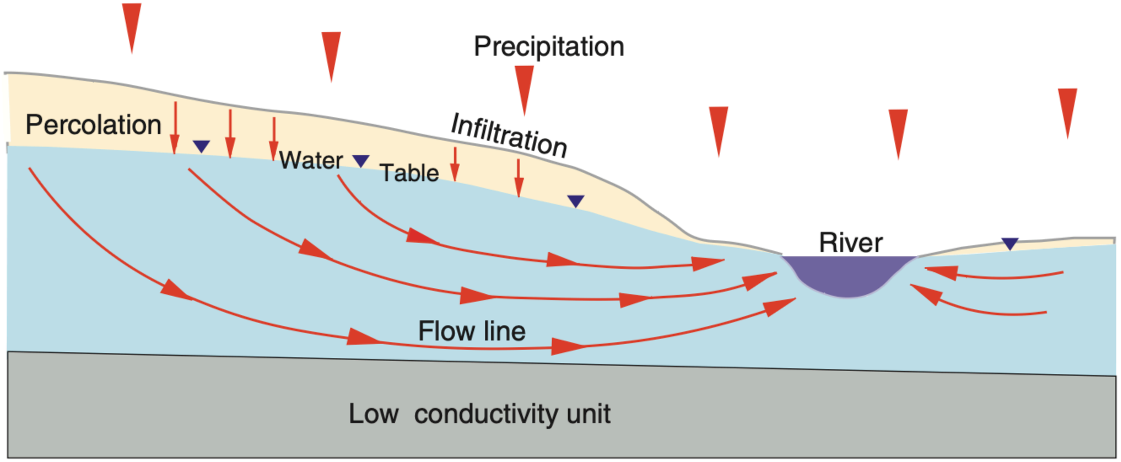

Figure 8: Groundwater flow system and discharge to stream

Flow is slow, deep, and essential to sustaining surface water.

Groundwater flow in the lab

Darcy’s Law: The Foundations

- **Henry Darcy** (1800s) studied water filtration and developed an empirical law for groundwater flow.
- Darcy’s experiments showed that **flow depends on energy differences**:
  - Water moves from **high to low gravitational energy**
  - Like a drop on a slide — higher up means more potential energy
- Energy loss over distance gives rise to the **hydraulic gradient**:

$$i = \frac{e_{top} - e_{bot}}{\Delta l}$$

- This principle forms the core of Darcy’s Law.

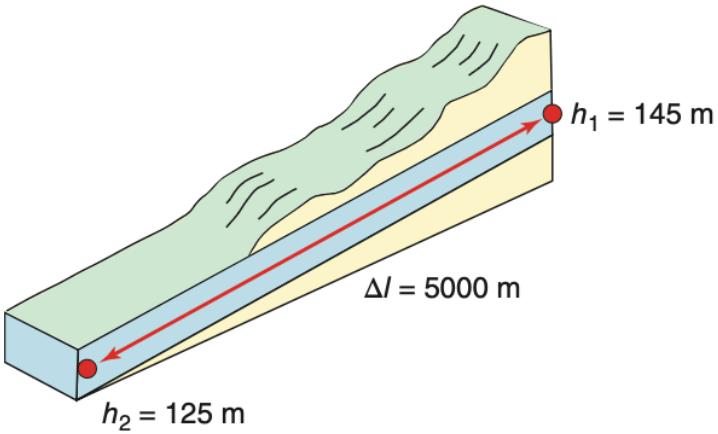

Figure 9: Water flows downhill due to gravitational energy

Hydraulic Gradient and Energy Slope

- **Gradient** = energy change per unit length of flow path
- In groundwater terms:
  - Energy is **hydraulic head** (gravitational + pressure)
  - Steeper gradient → faster potential flow
- Flow rate increases with:
  - Greater difference in head
  - Shorter flow distance
- Key analogy: **sliding down a playground slide**
    - Steeper slide → faster slide

The gradient controls the force pushing water through pores.

Figure 10: Gravitational energy slope analogy

Hydraulic Conductivity

- Not all materials transmit water equally well.
- **Hydraulic conductivity** (K) quantifies this:
  - High K → gravel, coarse sand
  - Low K → clay, unfractured rock
- Units: length per time (e.g. m/day)
- K depends on:
  - Pore size and connectivity
  - Fluid properties (viscosity, density)

Even with the same gradient, **some materials flow better than others**.

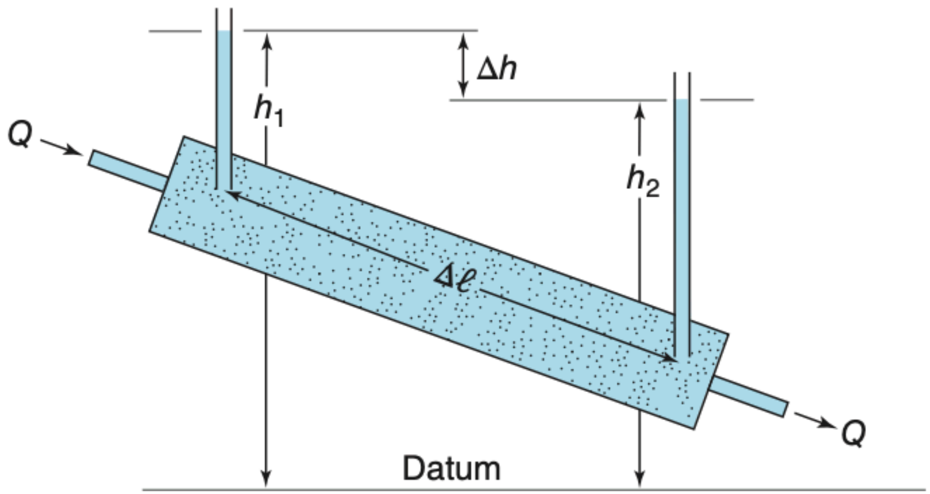

Figure 11: Darcy's column experiment setup

Darcy's Equation

- Darcy combined gradient (i) and material property (K) to describe flow:

  $$q = -Ki$$

  - q: Darcy velocity (aka specific discharge), volumetric flow per area

- Alternate form using cross-sectional area A:

  $$Q = -KiA$$

- Negative sign: flow goes from high head to low head
- Valid for **laminar flow** in porous media

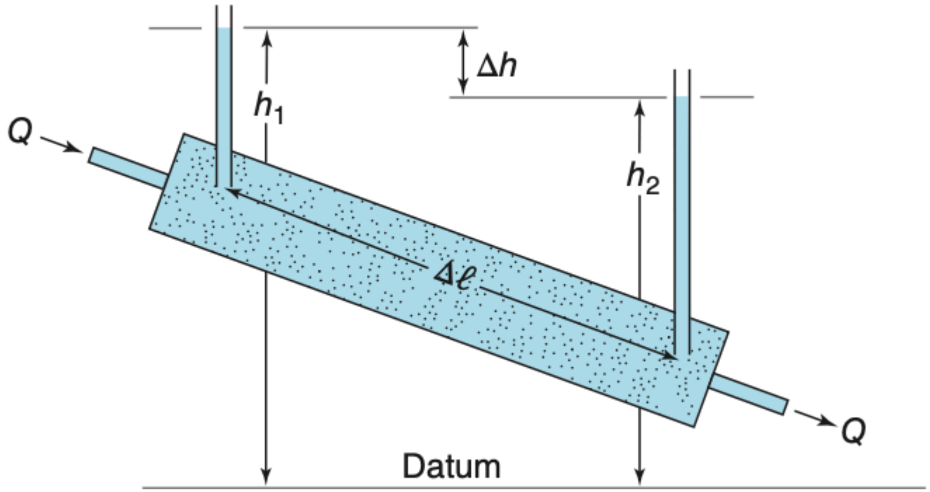

Figure 12: Q and h measured along a porous column

Darcy's Law is **foundational** to all groundwater flow modeling

Linear (Pore) Velocity

- **Darcy velocity** (q) assumes flow spreads over the whole cross section.
- But in reality, water only flows through **connected pores**
- True or **pore velocity** (v) is:

$$v = \frac{q}{n_e}$$

where:

  - n_e : effective porosity (only connected pore space)
  - q: Darcy velocity

- Since $n_e < 1$, pore velocity is **greater** than Darcy velocity
- Rearranged Darcy's Law:

$$v = -\frac{K}{n_e} \cdot \frac{h_1 - h_2}{\Delta l}$$

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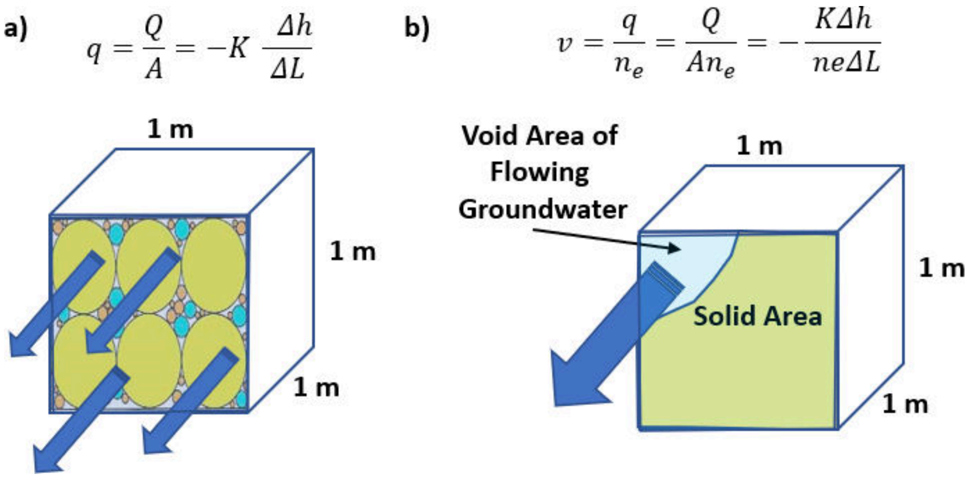


Figure 13: Water flows through effective pore space only.

This gives the **actual speed** water travels through aquifers.

Components of Hydraulic Head

- **Hydraulic head** is a measure of energy available to move groundwater
- It is expressed in units of **length** (e.g., m above sea level)
- In the field, head is measured with a **standpipe** or piezometer.
- Key components:
  - Elevation of top of casing (EL_{TOC})
  - Depth to water (DTW)
- The head is calculated by:

$$h = EL_{TOC} - DTW$$

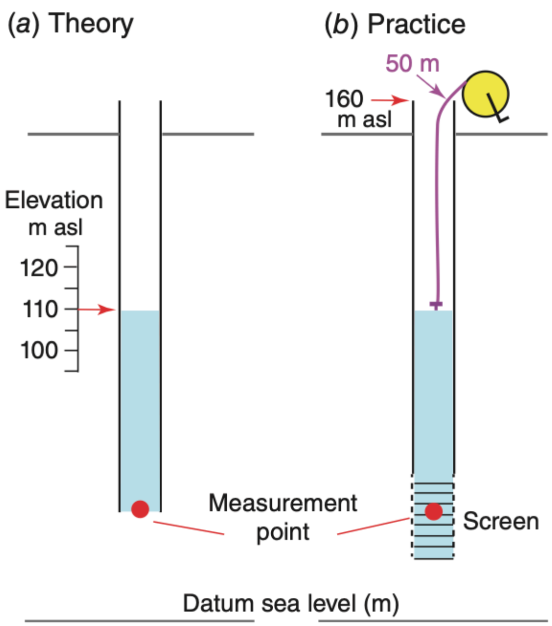


Figure 14: Measuring hydraulic head in practice and theory

Measuring head tells us the **direction and gradient** of groundwater flow.

Components of Hydraulic Head II

- Hydraulic head consists of:
  - **Elevation head** (z): height above datum
  - **Pressure head** ($P/\rho_w g$): height of water column exerting pressure
  - **Velocity head**: often negligible in groundwater
- General form (Bernoulli):

$$h = z + \frac{P}{\rho_w g} + \frac{v^2}{2g}$$

- Simplified form (no velocity term):

$$h = z + \frac{P}{\rho_w g}$$

- Units: meters (m), Pressure (Pa) can be converted from depth of water:

$$P = \rho_w g(h - z)$$

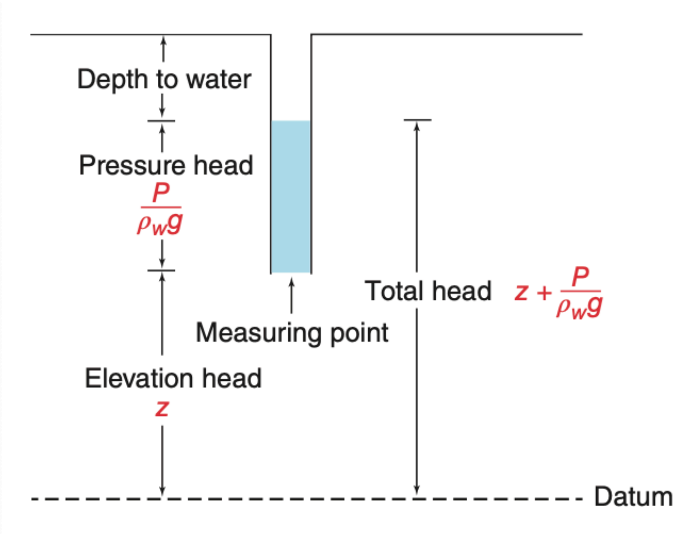


Figure 15: Total head = elevation + pressure head

Hydraulic Conductivity

- **Hydraulic conductivity** is a measure of how easily water flows through porous material
- Denoted by K, it appears in Darcy's Law:

$$q = -K\frac{dh}{dl}$$

- Units of K are velocity (e.g. m/day)
- $\uparrow K$: sand, gravel, $\downarrow K$: clay, shale
- K depends on:
  - Properties of the **medium**
  - Properties of the **fluid** (density, viscosity)

- **Intrinsic permeability** k characterizes the medium only:
  - Independent of fluid type
  - Units: m^2, also expressed in **darcy** or **cm²**
- Darcy's Law rewritten:

$$q = -\frac{k\,\rho_w\,g}{\mu}\,\frac{dh}{dl}$$

- Conversion to hydraulic conductivity:

$$K = \frac{k\,\rho_w\,g}{\mu}$$

- At 20°C and 1 atm: $ K = 9.77 \times 10^6 \cdot k$

$$k = (1.023) \times 10^{-7}\,\mathrm{m} \cdot \mathrm{sec} \cdot K$$

Hydraulic conductivity is widely used in field hydrogeology.

Estimating Hydraulic Conductivity

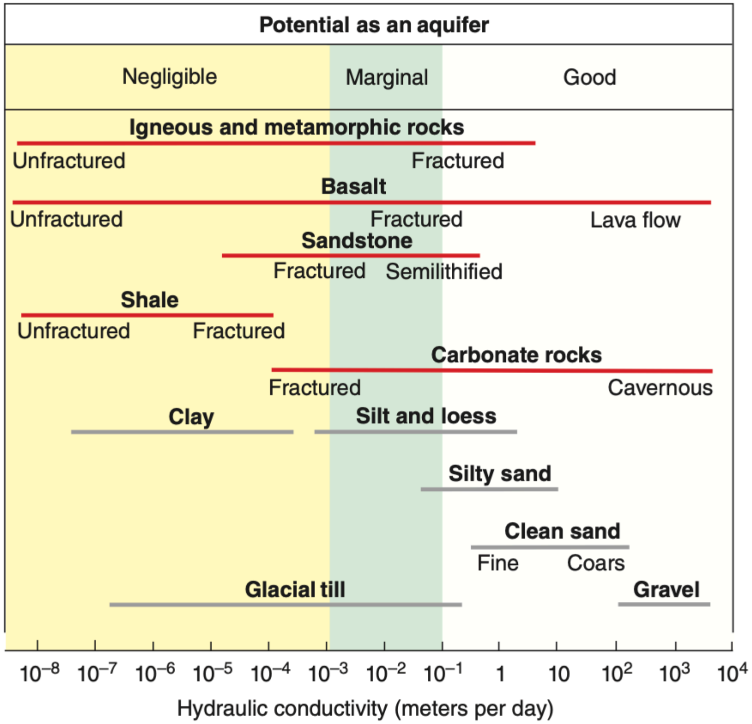


Figure 16: Hydraulic conductivity for selected acquifer materials

Source	Equation	Parameters
Hazen (1911)	$K = Cd_{10}^2$	K = hydraulic conductivity (cm/sec) C = constant 100 to 150 (cm/sec) for loose sand d_{10} = effective grain size cm (10% particles are finer, 90% are coarser)
Harleman et al. (1963)	$k = (6.54 \times 10^{-4})d_{10}^2$	k = permeability (cm^2)
Krumbein and Monk (1943)	$k = 760d^2e^{-1.31\sigma}$	k = permeability (darcys) d = geometric mean grain diameter (mm) σ = log standard deviation of the size distribution
Kozeny (1927)	$k = Cn^3/S^{*2}$	C = dimensionless constant: 0.5, 0.562, and 0.597 for circular, square, and equilateral triangle pore openings k = permeability (L^2) n = porosity S^* = specific surface-interstitial surface areas of pores per unit bulk volume of the medium
Kozeny-Carmen Bear (1972)	$K = \left(\frac{\rho_w g}{\mu}\right)\frac{n^3}{(1-n)^2}\left(\frac{d_m^2}{180}\right)$	K = hydraulic conductivity ρ_w = fluid density υ = fluid viscosity g = gravitational constant d_m = any representative grain size n = porosity

Figure 17: Examples of empirical relationships for estimating hydraulic conductivity or permeability values

Experimental Determination of Hydraulic Conductivity

Constant-Head Test

- Used for **coarse-grained** materials (e.g. sands, gravels)
- A steady flow is established by maintaining a **constant head difference** across the sample
- Flow rate Q is measured at the outlet
- Hydraulic conductivity K is calculated as:

$$K = \frac{QL}{Ah}$$

where:

  - L = sample length
  - A = cross-sectional area
  - h = constant head difference

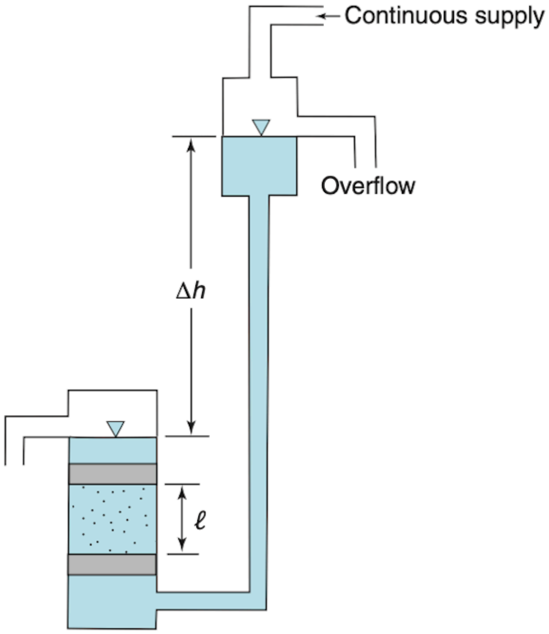


Figure 18: Constant-head permeameter

Works best when Q is large and easily measured.

Experimental Determination of Hydraulic Conductivity

Falling-Head Test

- Preferred for **fine-grained** or **less permeable** materials (e.g. clays, silts)
- No continuous flow: water level in standpipe drops over time
- Measure time $t_1 - t_0$ for water head to fall from h_0 to h_1
- Hydraulic conductivity:

$$K = 2.3 \frac{aL}{A(t_1 - t_0)} \log_{10} \left(\frac{h_0}{h_1} \right)$$

where:

  - a = standpipe cross-section
  - A = sample cross-section

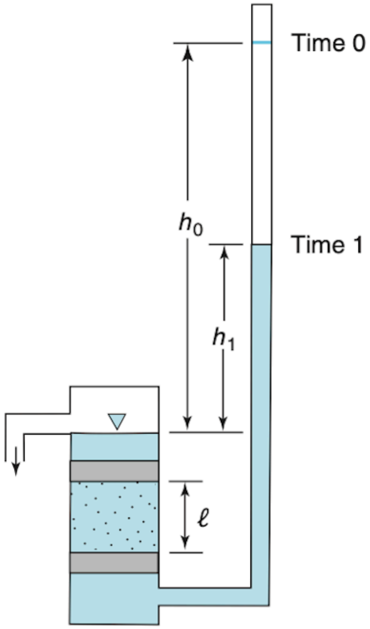


Figure 10: Falling-head permeameter

Falling-head is more accurate for slow flows and low K.

Anisotropic Flow and Darcy's Law

- In **anisotropic materials**, hydraulic conductivity depends on direction.
- Darcy's Law becomes a **vector-tensor** equation:

$$\mathbf{q} = -\mathbf{K}\nabla h$$

- In Cartesian form:

$$\mathbf{q} = q_x\,\mathbf{i} + q_y\,\mathbf{j} + q_z\,\mathbf{k}$$

$$\nabla h = \frac{\partial h}{\partial x}\mathbf{i} + \frac{\partial h}{\partial y}\mathbf{j} + \frac{\partial h}{\partial z}\mathbf{k}$$

- The hydraulic conductivity tensor is:

$$\mathbf{K} = \begin{bmatrix} K_{xx} & K_{xy} & K_{xz} \\ K_{yx} & K_{yy} & K_{yz} \\ K_{zx} & K_{zy} & K_{zz} \end{bmatrix}$$

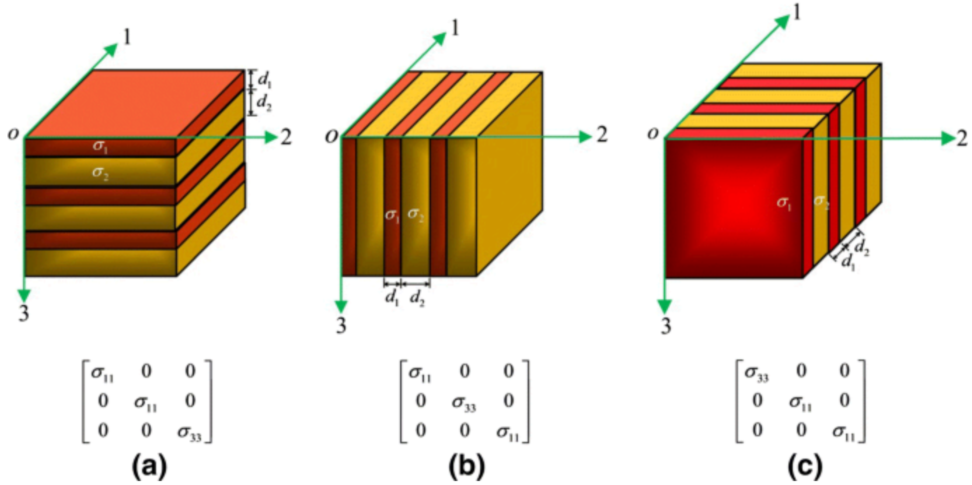


Figure 20: Anisotropic diffusivity makes material move with different velocities in different directions.

These terms describe how flow in one direction may be influenced by gradients in others.

Darcy Velocity Components

- For anisotropic flow, each component of flow is influenced by **multiple gradients**:

$$q_x = -K_{xx}\frac{\partial h}{\partial x} - K_{xy}\frac{\partial h}{\partial y} - K_{xz}\frac{\partial h}{\partial z}$$

$$q_y = -K_{yx}\frac{\partial h}{\partial x} - K_{yy}\frac{\partial h}{\partial y} - K_{yz}\frac{\partial h}{\partial z}$$

$$q_z = -K_{zx}\frac{\partial h}{\partial x} - K_{zy}\frac{\partial h}{\partial y} - K_{zz}\frac{\partial h}{\partial z}$$

- If the medium is **aligned with coordinate axes**, then off-diagonal terms vanish:

$$K_{xy} = K_{yx} = K_{xz} = K_{zx} = K_{yz} = K_{zy} = 0$$

- This simplifies to:

$$q_x = -K_{xx}\frac{\partial h}{\partial x}, \quad q_y = -K_{yy}\frac{\partial h}{\partial y}, \quad q_z = -K_{zz}\frac{\partial h}{\partial z}$$

Anisotropy in Sedimentary Rocks

- Field and lab studies confirm many rocks are **anisotropic**
- Horizontal conductivity is often **orders of magnitude** greater than vertical
- This affects:
  - Recharge and discharge zones
  - Aquifer drawdown patterns
  - Contaminant transport direction

Below are typical values from core samples:

Table – Anisotropic Conductivity of Sedimentary Rocks

Material	Hydraulic conductivity (m/s)	Vertical conductivity (m/s)
Anhydrite	10^{-14} to 10^{-12}	10^{-15} to 10^{-13}
Chalk	10^{-10} to 10^{-8}	5×10^{-11} to 5×10^{-9}
Limestone, dolomite	10^{-9} to 10^{-7}	5×10^{-10} to 5×10^{-8}
Sandstone	5×10^{-13} to 10^{-10}	2.5×10^{-13} to 5×10^{-11}
Shale	10^{-14} to 10^{-12}	10^{-15} to 10^{-13}
Salt	10^{-14}	10^{-14}

Table 3

Heterogeneous Hydraulic Conductivity

- A medium is **homogeneous** if permeability is constant within it.
- If permeability varies across space, it is **heterogeneous**.
- A common model: **layered medium** with different K_{hi}, K_{vi}, and thicknesses b_i
- For M horizontal layers, the **equivalent horizontal conductivity** is:

$$K_h = \frac{\sum_{i=1}^{M} b_i K_{hi}}{\sum_{i=1}^{M} b_i}$$

- This is a **thickness-weighted average**.
- Horizontal conductivity dominates in flow parallel to bedding.

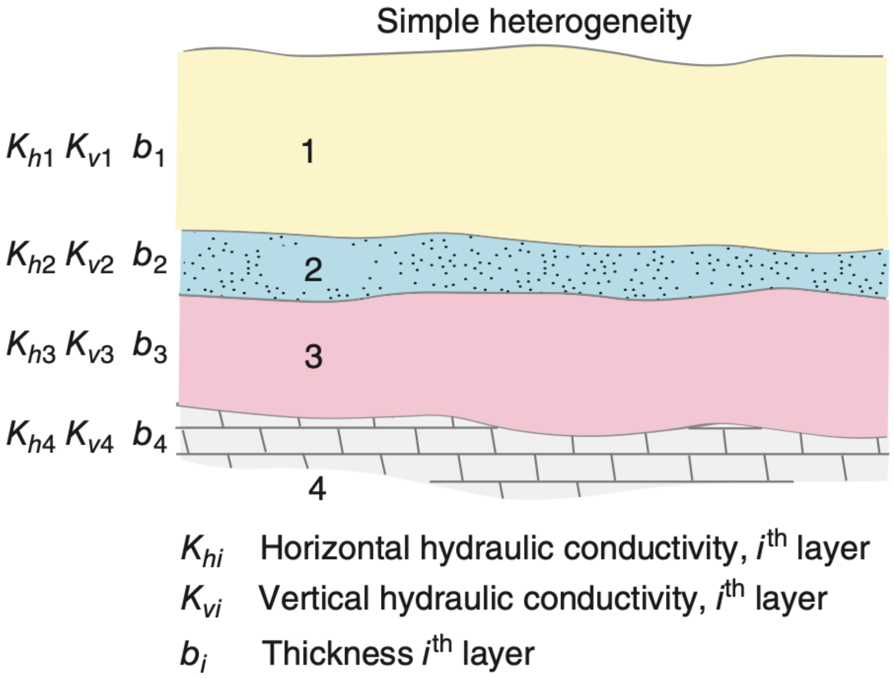


Figure 21: Layered heterogeneity with varying K_h, K_v

Vertical Flow and Flow Refraction

- In vertical direction, **equivalent conductivity** is given by:

$$K_v = \frac{\sum_{i=1}^{M} b_i}{\sum_{i=1}^{M} \frac{b_i}{K_{vi}}}$$

- This is a **harmonic average** — low K_{vi} dominates vertical resistance.
- When groundwater crosses a boundary between layers, it refracts.
- **Refraction law:**

$$\frac{K_1}{K_2} = \frac{\tan(\alpha_1)}{\tan(\alpha_2)}$$

- The angle of flow changes based on contrast in hydraulic conductivity across boundary.

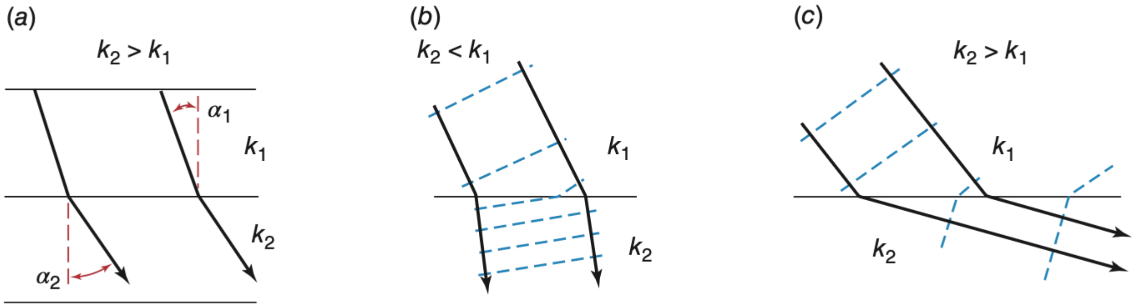


Figure 22: Flow refraction across geologic contacts

Investigating Groundwater Flow

- Hydraulic head can be measured with:
  1. Water wells
  2. Piezometers
  3. Water table observation wells
- These are often installed **together** at study sites
- Example system: deep aquifer beneath shallow, low-permeability unit
- Flow is:
  - Vertical in shallow layer
  - Horizontal in deeper aquifer

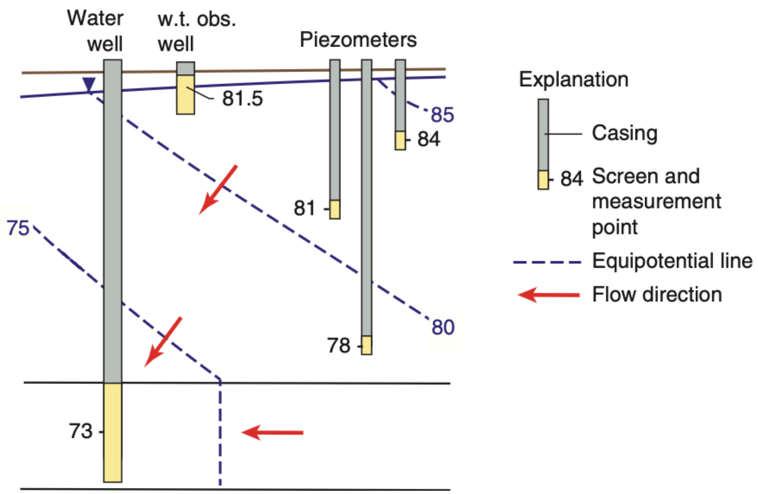
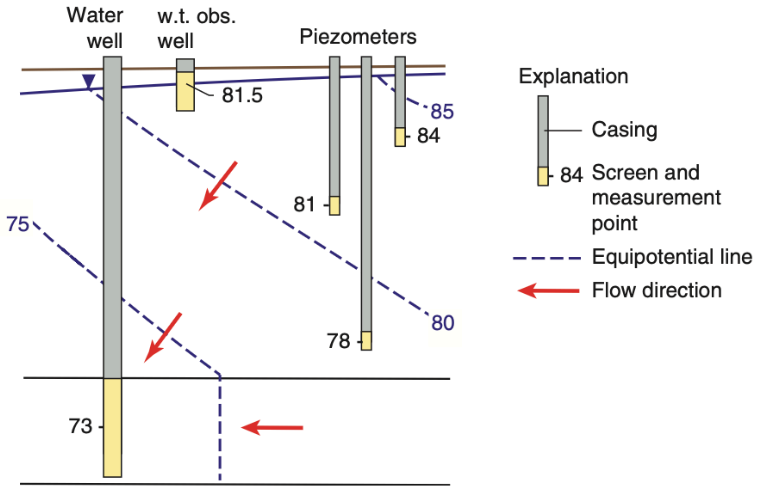


Figure 23: Wells, piezometers, and equipotential lines

Water Wells and Piezometers

- **Water wells**:
  - Large-diameter casings
  - Long screens → integrate over large depth
  - Used in well-defined aquifers
- Measure average head where equipotential lines are vertical
- In horizontal aquifer flow, screen depth doesn't affect head reading
- **Piezometers**:
  - Narrow diameter
  - Short screen → **point-specific** measurements
  - Ideal for contamination, dewatering, or vertical gradient studies
  - Often installed as **nests** at multiple depths

- Water well: $h = 73$ m (deep)
- Piezometers: show vertical gradient (e.g., 81 → 78 m)



Wells, piezometers, and equipotential lines

Potentiometric Surface Maps

- Mapping the spatial distribution of hydraulic heads in an aquifer is a key method for **evaluating aquifer conditions.**
- A **potentiometric surface map** shows groundwater levels (hydraulic head) across an area.
- Important uses:
  - Identifying **recharge and discharge areas**
  - Detecting **pumping impacts** (cones of depression)
  - Predicting **contaminant migration paths**
- **Meinzer (1923)**: defined potentiometric surface as the imaginary surface representing groundwater head in an aquifer.

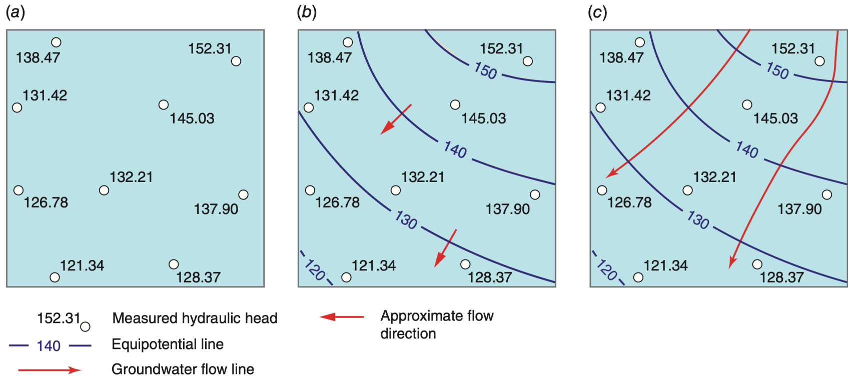


Figure 24: Steps in constructing a potentiometric surface map

Steps to Create Potentiometric Maps

- Three main steps:
    1. **Plot points**: map well/piezometer locations with measured head values
    2. **Contour**: draw equipotential lines connecting equal head values
    3. **Flow arrows**: add arrows perpendicular to equipotential lines (showing decreasing head)
- Key notes:
    - Map must be for **one aquifer only**
    - Flow in aquifers is assumed **horizontal**
    - Equipotential lines → vertical in cross-section, but projected to 2D on the map

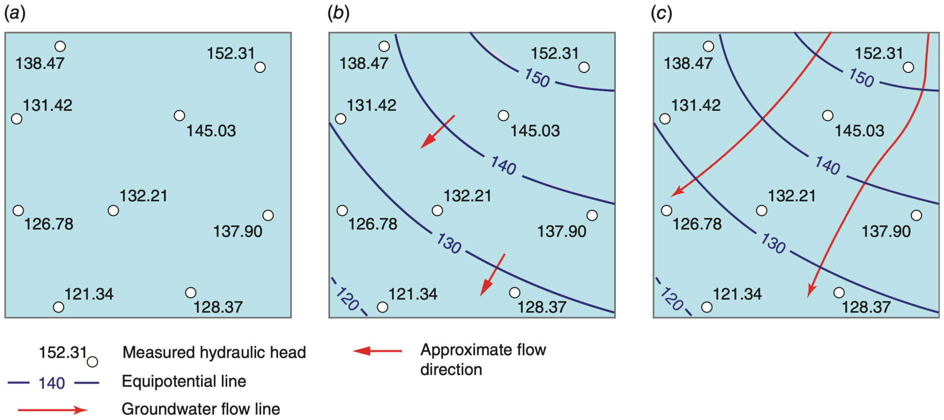


Figure 25: Equipotential lines and flow arrows on a potentiometric map

Example: Memphis Aquifer

- Example: **Memphis Sand Aquifer (1995)**
- Heavy pumping caused **cones of depression**
- Equipotential contours + flow arrows show water moving toward pumping centers
- **Applications**:
    - Tracking regional groundwater flow
    - Managing water supply stress
    - Identifying recovery trends after reduced pumping

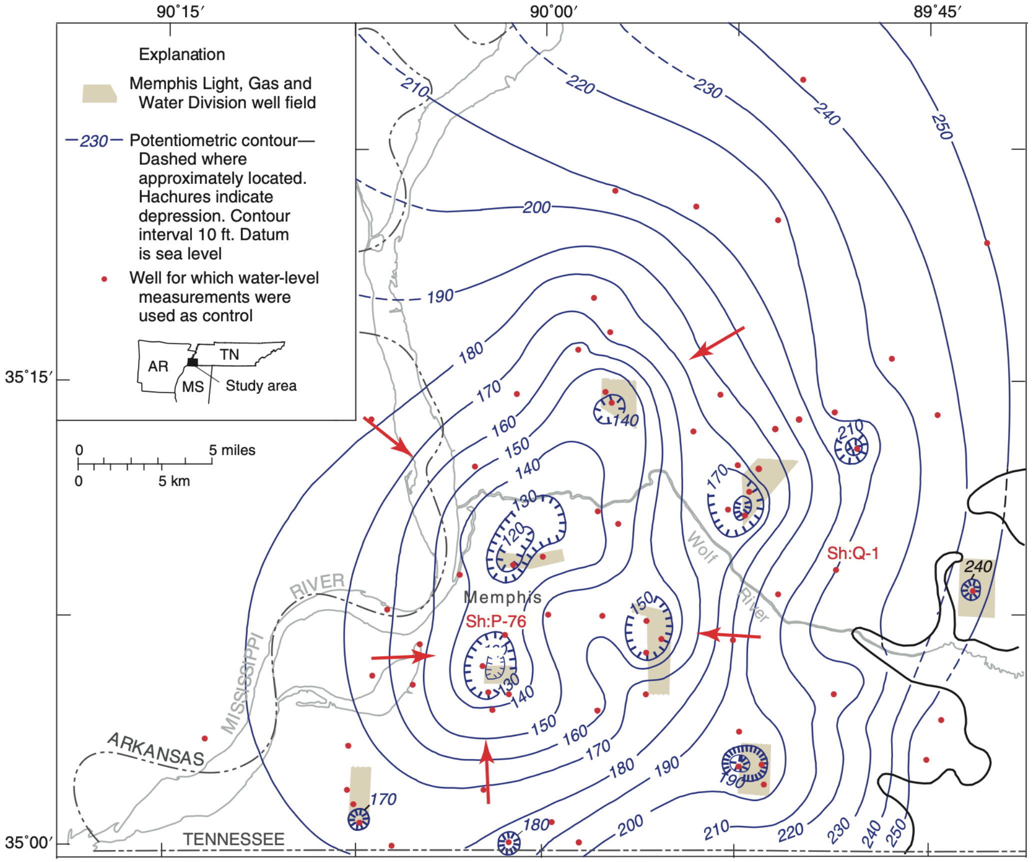


Figure 26: Potentiometric surface of the Memphis aquifer, showing cones of depression

Water-Level Hydrographs

- A **hydrograph** shows groundwater levels over time at a monitoring well
- Traditionally: manual measurements (monthly, seasonal)
- Now: automated, remote sensors (satellite uplinks, wireless)
- Hydrographs complement potentiometric maps:
  - Maps = snapshot at one time
  - Hydrographs = long-term temporal trends
- Useful for:
  - Detecting pumping impacts
  - Seasonal recharge/discharge cycles
  - Long-term aquifer depletion or recovery

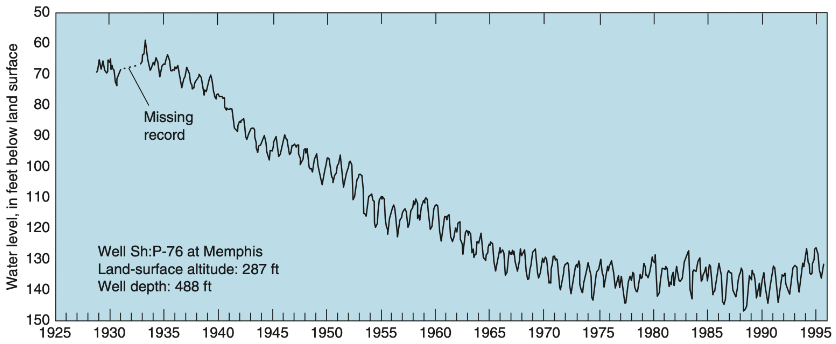


Figure 27: Long-term water-level hydrograph of Memphis aquifer well

Hydrogeological Cross Sections

- Vertical sections through aquifers showing:
  - **Stratigraphy** (layering, K contrasts)
  - **Hydraulic head distributions**
  - **Flow lines and equipotentials**
- Often constructed along **mean flow direction**
- Show how layering causes **refraction of flow paths**
- Vertical exaggeration helps visualize layering and flow

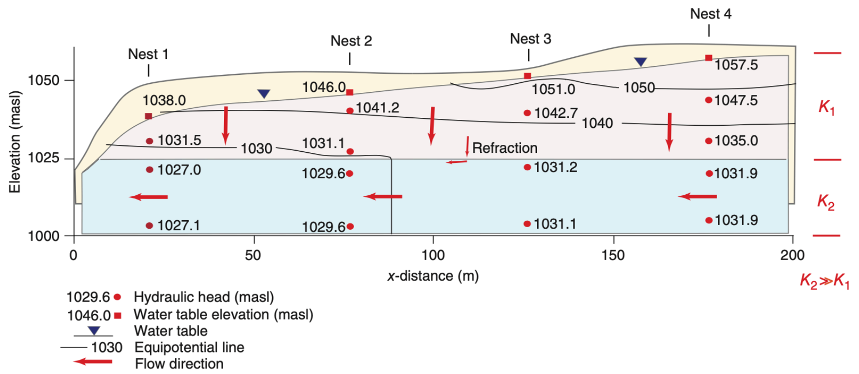


Figure 28: Hydrogeologic cross section with equipotential lines and flow refraction

What we learned

- We learned about the principles of groundwater flow, covering:
- **Porosity and Physical Properties**: void space in rocks and how it controls water storage
- **Groundwater Occurrence**: unsaturated vs saturated zones and the water table
- **Darcy's Law**: the fundamental equation governing groundwater flow rates
- **Hydraulic Properties**: head, conductivity, and permeability measurements
- **Heterogeneous Media**: flow behavior in layered systems and flow refraction
- **Field Investigation Methods**: wells, piezometers, and mapping techniques
- **Practical Applications**: aquifer management and contaminant transport